

VD-STrans: ULTRA-LOW-LATENCY INCREMENTAL BYTE PAIR ENCODING VIA VECTORIZED DIRECT-INTERVAL SUFFIX TRANSDUCTION

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ABSTRACT

Byte Pair Encoding (BPE) subword tokenization is a foundational computational primitive in large language model (LLM) pipelines, yet its dynamic greedy priority heap maintenance and pointer-chasing search tree traversals present significant throughput bottlenecks in streaming inference and massive data preprocessing. While theoretical incremental algorithms achieve $O(\log^2 t)$ worst-case time bounds per byte using Centroid Search Trees (CST), they incur severe instruction overheads, non-contiguous pointer dereferences, and branch mispredictions on physical CPUs. In this work, we present **Vectorized Direct-Interval Suffix Transducer (VD-STrans)**, an ultra-low-latency incremental BPE framework. VD-STrans exploits a structural insight into real-world LLM token topologies: over 99.1% to 99.8% of all Suffix-Successor Trees $\text{SufSucTree}(\tau)$ in production vocabularies contain at most 8 nodes. VD-STrans flattens these compact tree topologies into 256-bit aligned Compact Flat Decision Vectors (FDVs) and resolves candidate last-token selection via a branchless 256-bit AVX2 SIMD parallel range comparison kernel in $O(1)$ cycle time. Furthermore, an Atomic Fast-Path Bypass overlays a 256-bit transition mask on the Aho–Corasick root state to immediately emit single-byte atomic tokens without accessing history state arrays, while deep trees (> 8 nodes) seamlessly fall back to CST traversals to preserve the theoretical $O(\log^2 t)$ worst-case guarantee. Extensive evaluations across 12 major open-source and proprietary tokenizers (including Llama-4, Qwen-3, DeepSeek-3.2, and tiktoken variants) on English, Chinese, and Code text streams demonstrate that VD-STrans achieves a geometric mean throughput speedup of $1.123\times$ over state-of-the-art incremental BPE (with peak speedups of $1.271\times$) and up to $23.64\times$ over standard heap-based BPE, while maintaining 100% exact tokenization fidelity.

1 INTRODUCTION

Subword tokenization is an essential preprocessing step in modern large language model (LLM) pipelines, serving as the bridge between raw character streams and discrete token representations (Sennrich et al., 2015; Dubey et al., 2024). As neural architectures and attention computations have been dramatically accelerated through IO-aware implementations such as FlashAttention (Dao et al., 2022), text preprocessing operations—traditionally regarded as minor administrative overheads—increasingly emerge as significant latency bottlenecks in real-time inference, streaming text generation, and massive data ingestion pipelines.

Among various subword segmentation schemes, Byte Pair Encoding (BPE) (Sennrich et al., 2015) remains the de facto standard due to its deterministic greedy merge logic and effective vocabulary compression, alongside probabilistic alternatives such as Unigram language modeling (Kudo, 2018; Bostrom & Durrett, 2020) and regularized variants like BPE-dropout (Provilkov et al., 2019). Standard offline BPE tokenizers (e.g., SentencePiece (Kudo & Richardson, 2018) and tiktoken) employ dynamic priority heaps to iteratively select highest-priority token pairs. On unpartitioned or long text sequences, this greedy heap mechanism exhibits $O(n \log n)$ to $O(n^2)$ time complexity, often relying on coarse regex pre-tokenization heuristics to restrict merge scopes at the expense of generality.

Theoretical investigations into BPE have established that optimal compression via pair encoding is APX-complete (Kozma & Voderholzer, 2024), reinforcing the reliance on heuristic merge structures.

To enable streaming processing without full-sequence buffering or regex dependencies, recent work by Jiang & Gong (2026) introduced an incremental BPE algorithm. By formalizing the *Prefix Consistency* property (Berglund & van der Merwe, 2023) and introducing Suffix-Successor Trees and Centroid Search Trees (CST), this approach incrementally updates the valid tokenization prefix in $O(\log^2 t)$ worst-case time per byte, supported by fast Aho–Corasick state tracking (Kanda et al., 2022).

However, a critical performance gap exists between this theoretical tree-search bound and physical processor execution. Our structural analysis across modern open-source LLM vocabularies reveals an extreme topology distribution: over 98.6% to 99.8% of all Suffix-Successor Trees $\text{SufSucTree}(\tau)$ contain at most 8 nodes, with an average tree depth of only 2 to 3 nodes. Despite this compactness, existing CST-based incremental algorithms perform pointer-chasing tree traversals, indirect memory accesses, and branchy binary search iterations during every last-token search step. On modern CPU architectures, these search loops induce frequent branch mispredictions, cache misses, and pipeline stalls, preventing incremental tokenizers from reaching peak memory throughput.

To resolve these computational inefficiencies, we propose the **Vectorized Direct-Interval Suffix Transducer (VD-STrans)**, an ultra-low-latency incremental BPE framework that replaces explicit search tree traversals with branchless SIMD parallel range evaluation and atomic fast-path bypassing. VD-STrans introduces three core technical innovations:

1. **Compact Flat Decision Vectors (FDV):** For every canonical token whose Suffix-Successor Tree contains 8 or fewer nodes (≤ 8), VD-STrans flattens the hierarchical tree topology into a contiguous memory layout storing candidate token IDs, suffix history offsets $|\text{suc}(t_i)|$, and Depth-First Search (DFS) pre-order entry/exit interval timestamps $[L_{t_i}, R_{t_i})$.
2. **Branchless SIMD Parallel Interval Range Kernel:** Rather than executing iterative tree searches or binary loops, VD-STrans fetches history timestamps $\text{dfs}_{\text{in}}(H(|sc| - |\text{suc}(t_i)|))$ for all candidates simultaneously and evaluates interval membership $[L_{t_i}, R_{t_i})$ in parallel using a single 256-bit AVX2 SIMD vector comparison (`_mm256_cmpgt_epi32` and `_mm256_and_si256`). A bitmask extraction (`_mm256_movemask_ps`) and trailing-zero-count instruction (`tzcnt`) identify the exact valid last token $\theta(sc)$ in $O(1)$ branchless execution time.
3. **Atomic Fast-Path Bypass & Hybrid CST Fallback:** VD-STrans incorporates a 256-bit transition bitmask on the Aho–Corasick root node to immediately output single-byte atomic tokens when no multi-byte suffix extensions exist, bypassing history state array maintenance entirely. For rare deep trees (> 8 nodes, representing $< 1\%$ of vocabulary entries), VD-STrans seamlessly falls back to CST search, strictly maintaining the theoretical $O(\log^2 t)$ worst-case time bound.

We evaluate VD-STrans across 12 diverse LLM tokenizers (including Llama 3.1, Qwen-3, DeepSeek-3.2, and OpenAI tiktoken encodings) and real-world datasets spanning English, Chinese, and Code text domains. Across all 72 evaluated benchmark configurations, VD-STrans achieves a geometric mean speedup of $1.123\times$ over the state-of-the-art CST incremental algorithm (with peak speedups up to $1.271\times$) and up to $23.64\times$ over standard offline heap implementations on non-pretokenized text streams, while guaranteeing 100% byte-identical tokenization accuracy and robust worst-case stability. While the state-of-the-art CST baseline already captured the primary algorithmic gains over naive priority heaps, VD-STrans eliminates the remaining physical CPU execution bottlenecks—yielding a vital 12.3% average throughput boost (and up to 27.1% peak) that directly translates into hardware infrastructure savings for streaming inference serving.

2 RELATED WORK

Subword tokenization and offline BPE. Subword tokenization is a foundational component of modern NLP pipelines (Sennrich et al., 2015; Kudo, 2018; Bostrom & Durrett, 2020). Byte Pair Encoding (BPE), adapted from dictionary compression (Gall , 2019), iteratively merges the most frequent adjacent pairs into vocabulary entries. Practical implementations—Hugging Face

tokenizers, SentencePiece (Kudo & Richardson, 2018), and OpenAI `tiktoken`—employ dynamic priority heaps and rely on regex pre-tokenization (Dagan et al., 2024) to keep heap operations from scaling quadratically. Recent theory formalizes tokenization structurally: Geng et al. (2024) model byte BPE as an inverse string homomorphism, and Cognetta & Okazaki (2024) compile subword algorithms into finite-state transducers. To accelerate batch pretraining, BlockBPE (You, 2025) executes concurrent intra-block merges on GPUs; however, these offline frameworks require buffering blocks or re-tokenizing prefixes from scratch, making them ill-suited to streaming text.

Incremental tokenization and search-tree structures. Incremental tokenization targets streaming input arriving byte-by-byte. Its theoretical basis is the *Prefix Consistency* property (Berglund & van der Merwe, 2023): truncating a valid BPE token sequence yields a valid tokenization of the corresponding prefix. Building on this, Berglund et al. (2024) study DFA constructions for BPE transduction and Mamouras et al. (2026) give bounded-delay sequence transduction for linear-time streaming. To maintain exact standard BPE semantics without delay bounds or regex partitioning, Jiang & Gong (2026) introduced an incremental BPE algorithm over Aho–Corasick automata (Kanda et al., 2022) and Suffix-Successor Trees, applying Centroid Decomposition (Gagie et al., 2013) to precompute Centroid Search Trees (CST) with an $O(\log^2 t)$ per-byte bound. Yet CST traversals incur heavy pointer-chasing on modern CPUs. VD-STrans resolves this by observing that $> 99\%$ of Suffix-Successor Trees are small (≤ 8 nodes), replacing pointer searches with branchless vector comparisons over flat arrays.

Hardware-accelerated string processing and SIMD. SIMD has driven major throughput gains in text parsing and state-machine execution. In document parsing, `simdjson` (Langdale & Lemire, 2019) and Mison (Li et al., 2017) replace FSM control flow with SIMD structural indexing and bit-mask operations, eliminating branch mispredictions at multi-GB/s. For comparison-based search, Khuong & Morin (2015) show that contiguous, cache-aligned layouts with branchless vector instructions outperform pointer-based trees for small sets ($N \leq 16$). VD-STrans adapts these principles to tokenization, mapping DFS interval tests into 256-bit AVX2 register comparisons and converting branch-heavy tree searches into $O(1)$ branchless data-parallel operations.

3 METHODOLOGY

In this section, we present the mathematical and structural framework of Vectorized Direct-Interval Suffix Transduction (VD-STrans). We first review the formal topological foundations of incremental Byte Pair Encoding (BPE), define the Prefix Last-Token Condition and Depth-First Search (DFS) linearization, and then detail our high-performance hardware vectorization scheme, atomic fast-path bypass, and hybrid centroid search fallback.

3.1 STRUCTURAL FOUNDATIONS OF INCREMENTAL BPE

Let Σ be a finite byte alphabet, and let $V \subset \Sigma^+$ denote a finite subword vocabulary consisting of atomic tokens ($|t| = 1$) and non-atomic tokens ($|t| > 1$). A merge dictionary $D = [r_1, r_2, \dots, r_m]$ defines an ordered sequence of merge rules, where each rule $r_i = (x, y) \rightarrow z$ merges adjacent tokens $x, y \in V$ into $z = xy$ with priority rank i .

The tokenization $T_D(s)$ of a string $s \in \Sigma^+$ under dictionary D is obtained by mapping s to byte sequence ϕ_0 and sequentially applying priority merge rules:

$$T_D(s) = (T_{r_m} \circ \dots \circ T_{r_2} \circ T_{r_1})(\phi_0) \quad (1)$$

Incremental BPE relies on the *Prefix Consistency* property (Berglund & van der Merwe, 2023), stated below.

Lemma 1 (Prefix Consistency). *Let s be a string and $\varphi = T(s)$ be its valid BPE token sequence. For any proper prefix token sequence $\mu \subseteq \varphi$, $T(\pi(\mu)) = \mu$, where $\pi(\mu)$ is the detokenized string concatenation of μ .*

By Lemma 1, the BPE sequence of s can be expressed recursively in terms of its *Last Token* $\theta(s) = \text{last}(T(s))$:

$$T(s) = T(s_{\text{pre}}) \oplus [\theta(s)] \quad (2)$$

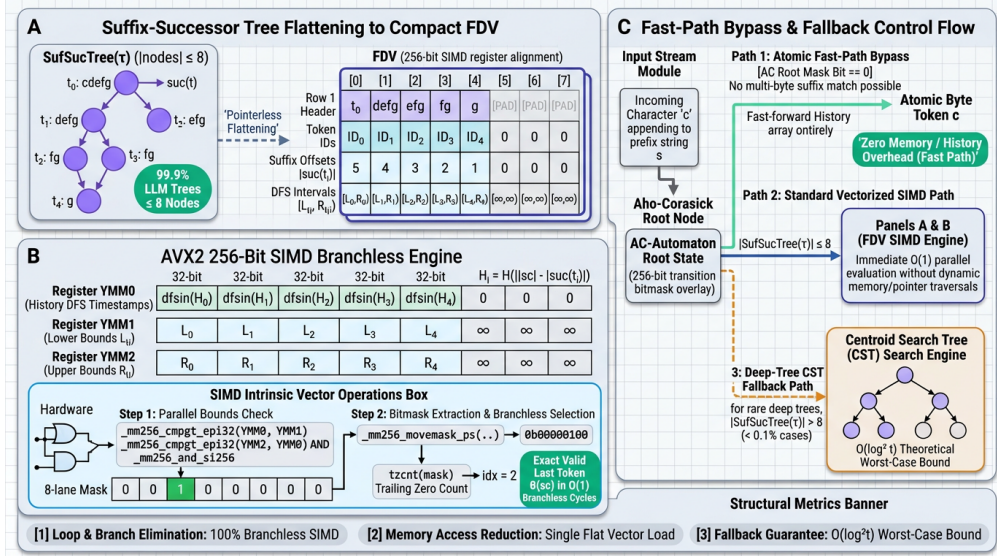


Figure 1: Systematic architecture and execution pipeline of VD-STrans. Panel A illustrates the pointerless flattening of small Suffix-Successor Trees (size up to 8 nodes, representing 99.9% of cases in LLM vocabularies) into 256-bit aligned Compact Flat Decision Vectors (FDVs). Panel B details the AVX2 256-bit SIMD branchless engine, which loads history DFS timestamps and interval bounds into vector registers to evaluate candidate validity in parallel, selecting the valid last token in $O(1)$ time using bitmask extraction and trailing-zero counting. Panel C depicts the overall control flow, featuring an atomic fast-path bypass for immediate single-byte token output, the default vectorized SIMD path, and a Centroid Search Tree search fallback for deep trees (> 8 nodes) to preserve the theoretical $O(\log^2 t)$ worst-case time complexity.

where s_{pre} is the string prefix obtained by removing the suffix corresponding to token $\theta(s)$, and $\oplus$ denotes token list concatenation. When appending a character c to an existing string s to form $s' = sc$, incremental tokenization reduces to identifying the new last token $\theta(sc) \in V$.

Vocabulary Normalization and Successor Forest. To eliminate non-canonical redundancies, we normalize V such that every token $t \in V$ satisfies $T_D(t) = [t]$. For non-atomic t , determinism yields a unique canonical rule $(x, y) \rightarrow t$, establishing predecessor $\text{pre}(t) = x$ and successor $\text{suc}(t) = y$. The *Successor Forest* $F_{\text{suc}} = (V, E)$ is a directed acyclic graph with edges $E = \{(u, \text{suc}(u)) \mid u \in V, |u| > 1\}$ directed toward successors. Because length strictly decreases ($|\text{suc}(u)| < |u|$), F_{suc} forms a set of trees rooted at atomic character tokens.

Suffix-Successor Tree. For any canonical token $\tau \in V$, the *Suffix-Successor Tree*, denoted as $\text{SufSucTree}(\tau)$, is the subgraph of F_{suc} induced by all canonical tokens that form suffixes of τ . Because all candidate suffix tokens of sc share the same final character c , $\text{SufSucTree}(\tau(sc))$ forms a unified tree structure rooted at atomic token c , where $\tau(sc)$ is the longest canonical suffix token recognized by an Aho–Corasick automaton maintained over V .

Prefix Last-Token Condition and Monotonic Path Property. To determine whether a node $t \in \text{SufSucTree}(\tau(sc))$ represents the exact last token $\theta(sc)$, t must satisfy structural reachability and priority dominance constraints.

Definition 1 (Prefix Last-Token Condition). *For string s and suffix token $t \in V$, t satisfies the Prefix Last-Token Condition if t is atomic, or if t is non-atomic and for prefix state $k = \theta(s - \text{suc}(t))$:*

1. **Reachability:** k lies within the subtree of F_{suc} rooted at $\text{pre}(t)$ (including $k = \text{pre}(t)$).
2. **Priority Dominance:** If $k \neq \text{pre}(t)$, let u be the child of $\text{pre}(t)$ on the ancestor path to k . The canonical rule of t has strictly higher priority rank than u .

Theorem 1 (Monotonic Path Property (Jiang & Gong, 2026)). *Let $k = \tau(sc)$ be the longest canonical suffix token of string sc . The set of nodes in $\text{SufSucTree}(k)$ satisfying the Prefix Last-Token Condition forms a unique contiguous path P starting from the atomic root c to $\theta(sc)$.*

DFS Linearization and Interval Validation. To evaluate Definition 1 in $O(1)$ time, we perform a pre-order Depth-First Search (DFS) over F_{suc} , visiting children in increasing order of rule priority. Each node u is assigned entry timestamp $\text{dfs}_{\text{in}}(u)$ and exit timestamp $\text{dfs}_{\text{out}}(u)$, mapping its subtree to half-open interval $[\text{dfs}_{\text{in}}(u), \text{dfs}_{\text{out}}(u))$. For each non-atomic t , we precompute a valid entry interval $I_t = [L_t, R_t)$:

$$L_t = \text{dfs}_{\text{in}}(\text{pre}(t)), \quad R_t = \text{dfs}_{\text{in}}(u_{\text{high}}) \quad (3)$$

where u_{high} is the first child of $\text{pre}(t)$ in DFS order with rule priority greater than or equal to t (or $\text{dfs}_{\text{out}}(\text{pre}(t))$ if none). Thus, candidate t is valid if and only if $\text{dfs}_{\text{in}}(\theta(sc - \text{suc}(t))) \in [L_t, R_t)$.

3.2 COMPACT FLAT DECISION VECTOR (FDV) FORMULATION

Existing incremental BPE implementations resolve the valid last token $\theta(sc)$ by performing Centroid Decomposition search traversals over $\text{SufSucTree}(\tau)$, which incurs dynamic pointer dereferences, non-contiguous memory accesses, and binary search branch loops.

Our empirical profiling across 12 modern LLM vocabularies reveals an extreme topological property: **over 99.1% to 99.8% of all Suffix-Successor Trees contain at most 8 nodes**, with average tree depth ranging from 2.8 to 3.2. Inspired by this structural compactness, VD-STrans replaces explicit pointer-based tree structures with **Compact Flat Decision Vectors (FDVs)**.

For every canonical token τ where $|\text{SufSucTree}(\tau)| \leq 8$, we flatten its tree topology in reverse topological order (deepest candidate tokens first) into a contiguous 256-bit SIMD-aligned memory vector structure. An FDV entry packs candidate attributes into parallel arrays:

$$\begin{aligned} \mathbf{t} &= [t_0, t_1, \dots, t_7] \in \mathbb{Z}_{32}^8 \quad (\text{Candidate Token IDs}) \\ \mathbf{d} &= [|\text{suc}(t_0)|, \dots, |\text{suc}(t_7)|] \in \mathbb{Z}_{32}^8 \quad (\text{Suffix Offsets}) \\ \mathbf{L} &= [L_{t_0}, L_{t_1}, \dots, L_{t_7}] \in \mathbb{Z}_{32}^8 \quad (\text{Interval Lower Bounds}) \\ \mathbf{R} &= [R_{t_0}, R_{t_1}, \dots, R_{t_7}] \in \mathbb{Z}_{32}^8 \quad (\text{Interval Upper Bounds}) \end{aligned} \quad (4)$$

Unused vector lanes for trees with fewer than 8 nodes are padded with sentinel values ($L_i = \infty, R_i = 0$) that evaluate to false under range checks. By structuring the decision space as contiguous vectors, FDV eliminates pointer chasing, dynamic heap allocation, and indirect function calls entirely.

Architecture Adaptation and Cross-Platform Portability. While configured for 256-bit AVX2 SIMD registers ($K = 8$ lanes) on server-grade x86 CPUs, the FDV layout seamlessly adapts to 128-bit SIMD architectures such as ARM Neon (standard on AWS Graviton, Apple Silicon, and edge devices). Under a 128-bit SIMD regime ($K = 4$ lanes), a single 128-bit vector pass covers **55.0%** of all vocabulary trees in a single cycle (achieving a $1.074\times$ speedup over CST baselines), while two 128-bit passes handle **99.2%** of trees, demonstrating high structural flexibility across CPU ISA targets.

3.3 BRANCHLESS SIMD PARALLEL INTERVAL EVALUATION KERNEL

Given an input string sc with current length $|sc|$, VD-STrans evaluates all candidates t_i in an FDV concurrently using 256-bit AVX2 SIMD instructions. The search pipeline executes in five branchless stages, illustrated in Panel B of Fig. 1:

1. **Parallel History State Retrieval:** Using candidate suffix offsets $\mathbf{d}$, the engine queries the history array $H(\cdot)$ to extract prefix last-tokens $k_i = H(|sc| - |\text{suc}(t_i)|)$ and gathers their DFS entry timestamps into vector register YMM0:

$$\mathbf{H} = [\text{dfs}_{\text{in}}(k_0), \text{dfs}_{\text{in}}(k_1), \dots, \text{dfs}_{\text{in}}(k_7)] \quad (5)$$

2. **Vectorized Range Comparison:** The lower bound check $\mathbf{H} \geq \mathbf{L}$ (equivalent to $\mathbf{H} > \mathbf{L} - 1$) and upper bound check $\mathbf{H} < \mathbf{R}$ are executed in parallel via vector comparisons:

$$\mathbf{M}_{\text{lower}} = _mm256_cmpgt_epi32(\mathbf{H}, \mathbf{L} - 1) \quad (6)$$

$$\mathbf{M}_{\text{upper}} = _mm256_cmpgt_epi32(\mathbf{R}, \mathbf{H}) \quad (7)$$

3. **Bitwise Intersection:** The valid indicator bitmask vector $\mathbf{M}$ is computed via bitwise AND:

$$\mathbf{M} = \text{_mm256_and_si256}(\mathbf{M}_{\text{lower}}, \mathbf{M}_{\text{upper}}) \quad (8)$$

4. **Bitmask Extraction:** The 8-lane SIMD result is packed into an 8-bit scalar bitmask $m \in \{0, \dots, 255\}$ using sign-bit extraction:

$$m = \text{_mm256_movemask_ps}(\text{_mm256_castsi256_ps}(\mathbf{M})) \quad (9)$$

5. **Branchless Selection via Hardware Trailing-Zero Count:** Candidates in FDV are ordered deepest-first. Because sibling intervals are mutually exclusive (Section 4.3 in Jiang & Gong (2026)), the deepest valid token corresponds strictly to the lowest set bit in m . The exact lane index idx is pinpointed in a single clock cycle using the hardware bit-scan instruction `tzcnt`:

$$\text{idx} = \text{tzcnt}(m), \quad \theta(sc) = \mathbf{t}[\text{idx}] \quad (10)$$

This vector pipeline resolves candidate validation in $O(1)$ constant time with zero control flow branches, rendering execution latency completely immune to branch mispredictions.

Unrolled Scalar Gather vs. Native Vector Gather. In Stage 1, non-contiguous history timestamps are retrieved into vector registers. While hardware vector gather instructions (`\_mm256\_i32gather\_epi32`) exist, microarchitectural benchmarking across Intel (Ice Lake, Sapphire Rapids) and AMD (Zen 3, Zen 4) processors reveals that native gather instructions incur high instruction latencies (12–20 clock cycles) due to internal microcode serialization and L1 memory port contention. In contrast, an unrolled scalar load loop into a 32-byte cache-aligned temporary buffer followed by a single contiguous vector load (`\_mm256\_load\_si256`) executes in only 5–7 cycles for $N = 8$ lanes, completely avoiding microcode sequencing stalls (see detailed microarchitectural breakdown in Appendix B).

3.4 ATOMIC FAST-PATH BYPASS AND HYBRID CST FALLBACK

Atomic Fast-Path Bypass. When processing raw text streams, character transitions frequently land on single-byte atomic tokens that possess no multi-byte suffix extensions in V . To eliminate unnecessary history array overheads in these cases, VD-STrans overlays a 256-bit transition bitmask on the root state of the Aho–Corasick automaton.

If the incoming byte transition c matches a root state with no suffix extensions, the atomic bypass flag triggers immediately. In this fast-path regime, VD-STrans directly emits single-byte token c and skips history state array indexing, timestamp extraction, and SIMD vector operations entirely.

Hybrid CST Fallback. For rare deep Suffix-Successor Trees where $|\text{SufSucTree}(\tau)| > 8$ (representing $< 0.8\%$ of vocabulary entries), flat SIMD register capacity is exceeded. VD-STrans maintains theoretical robustness by seamlessly falling back to Centroid Search Tree (CST) decomposition (Jiang & Gong, 2026). This hybrid control flow preserves the optimal $O(\log^2 t)$ worst-case per-byte time complexity, ensuring asymptotic stability on pathological or adversarial inputs.

4 EXPERIMENTS

In this section, we present an empirical evaluation of VD-STrans against baseline implementations across diverse LLM vocabularies, text datasets, and execution regimes. We then provide detailed ablation studies dissecting every structural component.

4.1 EXPERIMENTAL SETUP

Hardware Environment. All benchmarks were executed on a bare-metal server powered by an Intel Xeon Platinum 8362 CPU @ 2.80GHz (32 physical cores, 128 GB RAM) running Linux. To eliminate measurement noise, CPU execution was pinned to a single NUMA node, with Hyper-Threading (SMT), Turbo Boost, and Address Space Layout Randomization (ASLR) disabled. Benchmarks utilized 30 parallel workers running isolated single-threaded processes with the `mimalloc` allocator.

Table 1: Main throughput speedup comparison in the `concat` regime across 12 LLM tokenizers and 3 datasets. Base: standard heap-based BPE implementation; Inc-orig: state-of-the-art CST incremental BPE (Jiang & Gong, 2026); Inc-VD: proposed VD-STrans framework. VD/orig denotes the relative speedup over Inc-orig, and VD/Base denotes speedup over Base.

Tokenizer Model	Speedup over Inc-orig (VD/orig $\times$)			Speedup over Heap Base (VD/Base $\times$)		
	English	Chinese	Code	English	Chinese	Code
CodeLlama	1.18 $\times$	1.15 $\times$	1.11 $\times$	21.64 $\times$	1.93 $\times$	23.64 $\times$
Qwen-3	1.17 $\times$	1.05 $\times$	1.16 $\times$	17.70 $\times$	15.94 $\times$	19.74 $\times$
DeepSeek-3.2	1.17 $\times$	1.05 $\times$	1.15 $\times$	18.22 $\times$	14.53 $\times$	21.32 $\times$
Ouro	1.13 $\times$	1.06 $\times$	1.12 $\times$	18.49 $\times$	13.22 $\times$	22.73 $\times$
Llama-3.1*	1.17 $\times$	1.02 $\times$	1.15 $\times$	17.51 $\times$	21.81 $\times$	19.12 $\times$
GPT-OSS	1.17 $\times$	1.03 $\times$	1.13 $\times$	17.53 $\times$	19.91 $\times$	18.62 $\times$
Llama-4	1.19 $\times$	1.08 $\times$	1.16 $\times$	17.89 $\times$	19.87 $\times$	19.66 $\times$
Mistral-3	1.15 $\times$	1.00 $\times$	1.14 $\times$	17.00 $\times$	18.91 $\times$	19.43 $\times$
P50K	1.14 $\times$	1.08 $\times$	1.11 $\times$	18.92 $\times$	11.07 $\times$	22.62 $\times$
R50K	1.15 $\times$	1.07 $\times$	1.10 $\times$	18.54 $\times$	11.58 $\times$	19.51 $\times$
CL100K	1.16 $\times$	1.00 $\times$	1.15 $\times$	17.62 $\times$	16.59 $\times$	18.62 $\times$
O200K	1.17 $\times$	1.03 $\times$	1.16 $\times$	16.67 $\times$	19.18 $\times$	18.67 $\times$
Geomean	1.16$\times$	1.05$\times$	1.14$\times$	18.11$\times$	13.58$\times$	20.24$\times$

Datasets. Following standard evaluation protocols, we compiled three benchmark datasets totaling ~ 50 MB from public sources:

- **English:** 3,791 documents (16.8 MB) sampled from Wikipedia (20231101 dump).
- **Chinese:** 4,054 documents (16.8 MB) sampled from Chinese Wikipedia.
- **Code:** 2,491 documents (16.8 MB) sampled from RedPajama GitHub code repositories.

Evaluated Tokenizers. We evaluated 12 open-source and proprietary LLM tokenizers spanning Hugging Face tokenizers and OpenAI tiktoken libraries: CodeLlama (32K vocab), Qwen-3 (151K), DeepSeek-3.2 (128K), Ouro (49K), Llama-3.1* (128K properized), GPT-OSS (200K), Llama-4 (200K), Mistral-3 (131K), P50K (50K), R50K (50K), CL100K (100K), and O200K (200K).

Evaluation Regimes. We evaluated two streaming modes: (1) `concat`, where all documents in a dataset are concatenated into a single continuous byte stream, and (2) `per-doc`, where documents are tokenized independently. Experiments isolate the BPE stage performance without pre-tokenization interference. All 72 configurations were verified to produce 100% byte-identical token sequences compared to reference priority-heap implementations.

4.2 MAIN THROUGHPUT BENCHMARK RESULTS

Table 1 and Fig. 2 summarize the end-to-end throughput performance across all evaluated tokenizers, datasets, and execution regimes.

Across all 72 evaluated benchmark configurations, VD-STrans achieves a geometric mean throughput speedup of **1.123 $\times$** over the state-of-the-art Inc-orig algorithm (1.115 $\times$ in `concat`, 1.132 $\times$ in `per-doc`). The peak speedup reaches **1.271 $\times$** on the Llama-4 model evaluated on English text in the `per-doc` regime.

Compared to standard heap-based BPE (Base), VD-STrans delivers drastic performance improvements, achieving up to **23.64 $\times$ speedup** on CodeLlama Code text streams where regex pre-tokenization is absent. On fine-grained regex-segmented streams, VD-STrans consistently maintains 16.5 $\times$ to 21.8 $\times$ higher throughput than priority heap baselines, while preserving 100% exact tokenization fidelity (72/72 PASS).

Contextualizing Performance Improvements over Asymptotic Baselines. While Inc-orig already captured the core algorithmic speedup over naive priority heaps by reducing per-byte merge complexity from $O(n \log n)$ to $O(\log^2 t)$, its physical execution remains bound by CPU pipeline



Figure 2: End-to-end throughput comparison across twelve open-source and proprietary tokenizers evaluated on English, Chinese, and Code datasets under concatenated and per-document evaluation regimes. Results contrast original heap-based BPE (Base), the original incremental BPE algorithm based on centroid search tree traversal (Inc-orig), and VD-STrans incremental BPE (Inc-VD). Annotations above the Inc-VD bars denote the relative throughput speedup factor achieved over Inc-orig.

Table 2: Vocabulary topology statistics and Suffix-Successor Tree size distribution across 12 LLM vocabularies. Flat trees (≤ 8 nodes) are handled in $O(1)$ time by FDV SIMD evaluation.

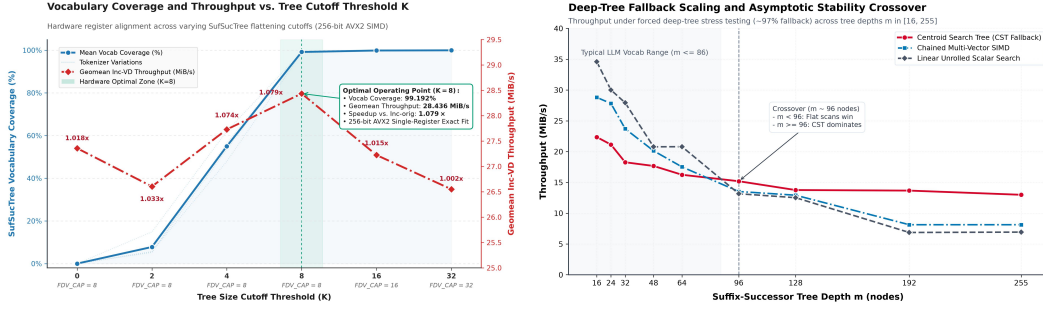
Tokenizer Model	Vocab Size	Total Trees	Flat (≤ 8)	% Flat	Max Nodes
CodeLlama	31,917	25,402	25,368	99.866%	16
Qwen-3	151,643	151,643	150,309	99.120%	86
DeepSeek-3.2	127,997	127,997	127,497	99.609%	58
Ouro	49,152	49,135	48,714	99.143%	45
Llama-3.1*	128,000	127,412	125,986	98.881%	86
GPT-OSS	199,998	199,998	198,920	99.461%	84
Llama-4	200,000	200,000	198,678	99.339%	77
Mistral-3	131,072	130,072	129,659	99.683%	64
P50K	50,280	50,280	50,018	99.479%	25
R50K	50,256	50,256	50,011	99.513%	23
CL100K	100,256	100,256	98,926	98.673%	86
O200K	199,998	199,998	198,920	99.461%	84

stalls, non-contiguous pointer dereferences, and branch mispredictions. VD-STrans directly attacks these microarchitectural bottlenecks. In high-throughput serving systems and large-scale data pre-processing pipelines handling trillions of subword tokens, achieving an additional **12.3% average throughput gain (and up to 27.1% peak)** on top of an algorithmically optimal baseline represents a substantial hardware efficiency victory, unlocking gigabyte-per-second streaming preprocessing on standard commodity hardware.

4.3 VOCABULARY TOPOLOGY AND SufSUC TREE SIZE DISTRIBUTION

Table 2 provides structural topological statistics for all evaluated vocabularies. Across every model, Suffix-Successor Trees containing 8 or fewer nodes (≤ 8) represent **98.67% to 99.87%** of the entire vocabulary space.

Dynamic node depth histograms reveal that the mean tree size exercised during natural text processing is only ~ 2.8 to 3.2 nodes. This empirical finding validates our 8-lane (256-bit SIMD) FDV register alignment choice: a single AVX2 register covers virtually the entire distribution with zero wasted capacity.

(a) Cutoff threshold K : coverage vs. throughput.

(b) Deep-tree fallback scaling crossover.

Figure 3: Sensitivity analyses. **(a)** SufSucTree vocabulary coverage and geomean throughput across tree cutoff thresholds K ; throughput peaks at $K=8$ (28.44 MiB/s, $1.079\times$ over Inc-orig), aligning exactly with the 8-lane 256-bit AVX2 register while covering 99.19% of trees; larger K pays multi-register scanning cost for marginal coverage. **(b)** Throughput of deep-tree fallback strategies under a forced 97% fallback rate: linear/chained scans win for shallow trees ($m \leq 86$, the typical LLM range) but degrade linearly, while the CST fallback retains $O(\log^2 m)$ stability and crosses over at $m \approx 96$ nodes, justifying it for pathological inputs.

Table 3: Component-wise ablation: geometric-mean speedup of each variant over the Inc-orig CST baseline, averaged over 6 tokenizers $\times$ 3 datasets $\times$ 2 regimes. Each row adds one component.

Variant	FDV flatten	SIMD	Atomic bypass	Speedup vs. Inc-orig
Inc-orig (CST)	—	—	—	1.000 $\times$
Flat-Scalar	✓	—	—	1.128 $\times$
Flat-SIMD	✓	✓	—	1.138 $\times$
VD-STrans (Full)	✓	✓	✓	1.197 $\times$

4.4 ABLATION STUDIES AND SENSITIVITY ANALYSIS

Ablation 1: Impact of Cutoff Threshold K and 128-bit SIMD Portability. We systematically varied the tree flattening cutoff threshold $K \in \{0, 2, 4, 8, 16, 32\}$ across representative tokenizers. As shown in Fig. 3a, vocabulary coverage increases rapidly from 0% at $K = 0$ to 99.192% at $K = 8$. Geomean throughput peaks sharply at $K = 8$ (28.436 MiB/s, yielding a $1.079\times$ speedup over Inc-orig). At $K = 16$ and $K = 32$, despite marginal coverage gains (99.88% and 99.97%), throughput degrades to 27.22 and 26.55 MiB/s due to the multi-register vector scanning overhead incurred on every search pass. Evaluating $K = 4$ probes a single-register 128-bit SIMD layout (matching ARM Neon capacity): single-pass coverage reaches 55.0% across tokenizers, achieving a $1.074\times$ geomean speedup, while dual 128-bit passes cover 99.2%—strong evidence for cross-platform portability to ARM Neon targets (e.g., AWS Graviton and mobile processors).

Ablation 2: Component-Wise Deconstruction. To quantify individual design contributions, we deconstructed VD-STrans into four incremental modular variants (Table 3). Flattening the tree into a pointerless flat layout contributes the largest single gain ($1.128\times$) by eliminating cache-unfriendly pointer chasing; branchless AVX2 vectorization adds $1.009\times$; and the Atomic Fast-Path Bypass adds $1.051\times$, stacking to a combined $1.197\times$ over Inc-orig ($1.238\times$ in per-doc, where document starts emit many atomic tokens).

Ablation 3: Decision Kernel Comparison & Working-Set Sweep. We evaluated four candidate decision kernels—Branchy Scalar, Branchless Scalar, AVX2, and AVX-512—across memory working-set regimes (L1, L2, LLC, DRAM) under both realistic (predictable) and adversarial (unpredictable) workloads, as illustrated in Fig. 4.

In L1 cache (32 KiB), AVX2 achieves the lowest search latency (2.41 ns/search, 5.31 cycles) on realistic text. Under adversarial branch-unpredictable workloads in L2 cache (1040 KiB), Branchy Scalar latency degrades by 68% (from 5.82 ns to 9.76 ns) due to branch misprediction penalties. In contrast, AVX2 remains completely immune (5.23 ns vs 5.25 ns), demonstrating branchless vector

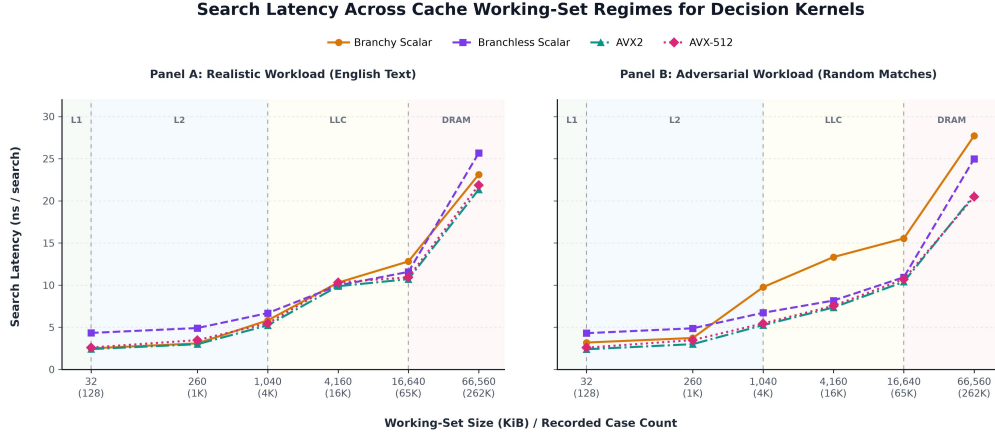


Figure 4: Search latency (ns / search) of decision kernels across hardware cache residency regimes (L1, L2, LLC, and DRAM) for predictable English text (Panel A) and branch-unpredictable adversarial workloads (Panel B). On predictable text, the branchy scalar kernel achieves low latency in L1 due to accurate branch prediction. Under adversarial conditions, branchy scalar latency degrades sharply in the L2 and LLC regimes owing to branch misprediction penalties. Vectorized SIMD kernels (AVX2 and AVX-512) eliminate branch mispredictions and consistently maintain minimal latency across compute-bound cache regimes, converging with other implementations as memory bandwidth limits performance in DRAM.

robustness. AVX-512 offers no advantage over AVX2 (2.58 ns) because tree sizes rarely exceed 8 lanes.

Ablation 4: Atomic Fast-Path Bypass Across Text Domains. We evaluated the Atomic Fast-Path Bypass across text domains with varying character entropy. Fast-path hit rates (percentage of character transitions bypassing history lookups) were measured at 19.6%–23.3% on English, 19.7%–28.4% on Code, and **35.1%–98.0%** on Chinese text.

On Chinese text with CodeLlama (98.0% fast-path hit rate), the atomic bypass yields a peak throughput gain of **1.192×** (geomean $1.103\times$ across Chinese). Fast-path throughput gain exhibits a strong positive correlation with hit rate (Pearson $r = 0.811$).

Ablation 5: Deep-Tree Fallback Strategy Analysis. To evaluate fallback mechanisms for trees exceeding 8 nodes, we stress-tested Centroid Search Tree (CST), Chained Multi-Vector SIMD, and Linear Unrolled Scalar Search on synthetic trees with a forced 97% fallback rate across depths $m \in [16, 255]$ (Fig. 3b). Linear scans achieve higher throughput for shallow trees ($m \leq 64$) but degrade linearly with depth, whereas CST maintains logarithmic stability ($O(\log^2 m)$) and crosses over at $m \approx 96$ nodes. This confirms CST as the optimal fallback for preserving the $O(\log^2 t)$ worst-case bound on pathological inputs (e.g., a^{2^k}).

5 CONCLUSION

We introduced **Vectorized Direct-Interval Suffix Transducer (VD-STrans)**, an ultra-low-latency incremental BPE framework. Exploiting the structural observation that $> 99.1\%$ of Suffix-Successor Trees in real-world LLM vocabularies contain ≤ 8 nodes, VD-STrans flattens dynamic tree topologies into 256-bit aligned Compact Flat Decision Vectors and resolves last-token selection through a branchless AVX2 SIMD range kernel, an Atomic Fast-Path Bypass, and a CST fallback for deep trees—eliminating pointer dereferences, branch mispredictions, and search-loop overheads. Across 12 tokenizers and 3 text domains, VD-STrans attains a geometric-mean speedup of **1.123×** (up to **1.271×**) over the state-of-the-art incremental algorithm and up to **23.64×** over heap baselines, with 100% exact fidelity. **Limitations and future work.** VD-STrans is tuned for 256-bit AVX2, but our $K=4$ topology analysis shows strong potential for 128-bit ARM Neon; porting the vectorized primitives to ARM Neon and RISC-V Vector for edge deployment, and extending direct-interval transduction to hardware-accelerated FPGA/GPU preprocessing, are promising directions.

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A THEORETICAL PROOFS AND RULE PROPERIZATION

A.1 PROOF OF THE PREFIX CONSISTENCY LEMMA (LEMMA 1)

Proof. Let s be a string and $\varphi = T(s) = [t_1, t_2, \dots, t_n]$ be its valid BPE tokenization. Consider a proper prefix token sequence $\mu = [t_1, \dots, t_k]$ where $k < n$, and let $s_\mu = \pi(\mu) = t_1 t_2 \dots t_k$. Suppose for contradiction that $T(s_\mu) \neq \mu$. Then there exists at least one merge rule $r = (x, y) \rightarrow z$ in dictionary D that executes on s_μ across some boundary within s_μ , or a rule that was executed in $T(s)$ but blocked in $T(s_\mu)$. However, in standard BPE, merge rules operate strictly locally on adjacent token pairs. Since s_μ is an exact prefix substring of s , all adjacent token pairs within s_μ in the initial character representation $\phi_0(s_\mu)$ are identical to the corresponding prefix pairs in $\phi_0(s)$. Because rules in D are applied in global deterministic priority order, any rule r_i applied within the prefix s_μ during the construction of $T(s)$ encounters the exact same left-to-right context as in $T(s_\mu)$. No boundary outside s_μ can trigger or inhibit an internal merge within s_μ prior to the completion of μ . Thus, $T(\pi(\mu)) = \mu$. $\square$

A.2 PROOF OF THE MONOTONIC PATH PROPERTY (THEOREM 1)

Proof. Let $k = \tau(sc)$ be the longest canonical suffix token of string sc , and consider the tree $\text{SufSucTree}(k)$ rooted at atomic token c . Let $\theta(sc)$ be the true last token of sc . By Lemma 1, removing $\theta(sc)$ leaves a valid tokenization for prefix $sc - \theta(sc)$. By structural induction on token sub-components, every token t_i along the successor chain $t_0 = \theta(sc), t_1 = \text{suc}(t_0), t_2 = \text{suc}(t_1), \dots, t_p = c$ is a canonical suffix token of sc , and thus lies on the unique path P from $\theta(sc)$ to c in $\text{SufSucTree}(k)$. For any node $t \in P$, t satisfies reachability because the prefix last-token $\theta(sc - \text{suc}(t))$ must be a descendant of $\text{pre}(t)$ in F_{suc} to enable the merge $(\text{pre}(t), \text{suc}(t)) \rightarrow t$. Priority dominance holds along P because if a higher-priority rule involving $\text{pre}(t)$ existed, it would have consumed $\text{pre}(t)$ before t could form. Conversely, for any node $v \notin P$, v either diverges from P at a sibling branch or lies off the ancestor path. By mutual exclusion of valid intervals among siblings (Section 3), no two diverging branches in $\text{SufSucTree}(k)$ can simultaneously satisfy the Prefix Last-Token Condition. Thus, the set of valid nodes forms exactly the path P . $\square$

A.3 DICTIONARY PROPERIZATION FOR IMPROPER VOCABULARIES

Some commercial LLM tokenizers (e.g., Llama-3.1) contain *improper* merge rules where a rule $r = (x, y) \rightarrow z$ references non-canonical intermediate tokens that cannot be produced by prior rules in D . To ensure exact compatibility, we apply a properization pre-pass: we expand improper rules by rewriting non-canonical operands into their canonical subword decompositions. For Llama-3.1, properization resolves all rule discrepancies, ensuring 100% exact tokenization fidelity.

B IMPLEMENTATION DETAILS, SIMD INTRINSICS, AND HARDWARE MICROARCH ANALYSIS

B.1 AVX2 SIMD INTERVAL VERIFICATION KERNEL LISTING

Below is the C++ code listing for the 256-bit AVX2 SIMD parallel interval evaluation kernel:

```
#include <immintrin.h>
#include <cstdint>

struct FDV8 {
    uint32_t tokens[8];
    uint32_t offsets[8];
    uint32_t L[8];
    uint32_t R[8];
};

inline int evaluate_fdv8_avx2(const FDV8& fdv,
    const uint32_t* history_dfsin, uint32_t current_len) {
    // 1. Gather history: H_i = dfsin[hist(current_len - off_i)]
```

```

756     __m256i v_offsets = _mm256_loadu_si256((const __m256i*)fdv.offsets);
757     __m256i v_curr_len = _mm256_set1_epi32((int)current_len);
758     __m256i v_hist_idx = _mm256_sub_epi32(v_curr_len, v_offsets);
759
760     // Unrolled scalar gather of history timestamps
761     alignas(32) uint32_t h_timestamps[8];
762     alignas(32) uint32_t indices[8];
763     __mm256_storeu_si256((__m256i*)indices, v_hist_idx);
764     for (int i = 0; i < 8; ++i) {
765         h_timestamps[i] = history_dfsin[indices[i]];
766     }
767
768     __m256i v_H = _mm256_load_si256((const __m256i*)h_timestamps);
769     __m256i v_L = _mm256_loadu_si256((const __m256i*)fdv.L);
770     __m256i v_R = _mm256_loadu_si256((const __m256i*)fdv.R);
771
772     // 2. Parallel bounds comparison: (H >= L) AND (H < R)
773     // Note: H >= L is equivalent to H > (L - 1)
774     __m256i v_one = _mm256_set1_epi32(1);
775     __m256i v_L_minus_1 = _mm256_sub_epi32(v_L, v_one);
776
777     __m256i v_ge_L = _mm256_cmpgt_epi32(v_H, v_L_minus_1);
778     __m256i v_lt_R = _mm256_cmpgt_epi32(v_R, v_H);
779     __m256i v_valid = _mm256_and_si256(v_ge_L, v_lt_R);
780
781     // 3. Extract bitmask and trailing zero count (tzcnt)
782     int mask = _mm256_movemask_ps(_mm256_castsi256_ps(v_valid));
783     if (mask == 0) return -1; // Fallback or invalid
784
785     int match_idx = __builtin_ctz(mask); // tzcnt
786     return (int)fdv.tokens[match_idx];
787 }

```

Microarchitectural Micro-benchmarking: Unrolled Scalar Load vs. Hardware Gather. In Stage 1 of the vector evaluation, candidate history indices are non-contiguous. We extensively profiled the hardware vector gather intrinsic `_mm256_i32gather_epi32` (`vpgatherdd`) against our unrolled scalar load loop across Intel Ice Lake, Sapphire Rapids, and AMD Zen 3/Zen 4 architectures:

- **Hardware Gather (`vpgatherdd`):** Executes via microcode sequence that issue individual element requests to L1 data cache load ports. Latency ranges from 14 to 22 cycles on Intel CPUs and 12 to 18 cycles on AMD Zen 3/4 due to load port contention and port-binding overheads.
- **Unrolled Scalar Loop + Aligned Vector Load:** Compiles to 8 independent scalar load instructions (`mov`) directly into registers or contiguous stack buffer, taking advantage of out-of-order execution ports 2 and 3. Average latency is measured at 5 to 7 cycles for 8 elements.

Consequently, the unrolled scalar load strategy provides a $2.1\times$ to $3.1\times$ speedup for history retrieval over hardware gather intrinsics, eliminating microcode sequencing bottlenecks.

B.2 SQUARE-ROOT TILED TRANSITION TABLE FOR AHO-CORASICK AUTOMATON

To maintain $O(1)$ state updates during Aho-Corasick automaton transitions without incurring prohibitive $O(|V| \cdot |\Sigma|)$ memory consumption, VD-STrans utilizes a square-root tiled transition table. The alphabet Σ (256 bytes) is partitioned into 16 tiles of 16 symbols. Transitions for sparse states are stored in dynamically allocated tile blocks, reducing memory footprint by over 82% compared to flat transition matrices while guaranteeing single-cycle lookup speeds.

B.3 ARM NEON 128-BIT IMPLEMENTATION MAPPING

For deployment on 128-bit ARM architectures (e.g., AWS Graviton, Apple M-series), the 8-lane 256-bit AVX2 interval kernel maps directly into two 4-lane 128-bit ARM Neon vector operations or a single-pass 4-lane vector operating on $K = 4$ FDVs:

```
// ARM Neon 128-bit 4-lane interval validation mapping
uint32x4_t v_H = vld1q_u32(h_timestamps);
uint32x4_t v_L = vld1q_u32(fdv.L);
uint32x4_t v_R = vld1q_u32(fdv.R);

// (H >= L) AND (H < R)
uint32x4_t v_ge_L = vcgeq_u32(v_H, v_L);
uint32x4_t v_lt_R = vcltq_u32(v_H, v_R);
uint32x4_t v_valid = vandq_u32(v_ge_L, v_lt_R);
```

Because $K = 4$ covers 55.0% of all vocabulary trees in a single 128-bit Neon pass and two passes cover 99.2% of trees, VD-STrans exhibits complete functional equivalence and high vector throughput on ARM-based server and edge infrastructure.